

Practical	MCA102P: Database Management Systems lab
Teaching Hours: 04 Hours/Week	Credits: 02
Duration of Exam: 03 Hours	Maximum Marks:50 (Exam 35 + IA 15)

PART A

1. Create and Manage University Database- Perform the following operations:

- Create tables: Student, Course, and Enrolment with appropriate attributes and constraints (Primary key, Foreign key).
- Insert minimum 5 records into each table.
- Display all student details enrolled in a specific course.
- Update a student's name using UPDATE command.
- Delete a course using DELETE command.

2. Perform the following operations:

- Create tables: Employee, Department, and Salary.
- Insert sample records into each.
- Display list of employees working in a specific department using JOIN.
- Display total salary paid per department using GROUP BY and aggregate function.
- Create a view that displays employee name, department, and salary.

3. Perform the following operations:

- Create tables: Books, Authors, Borrowers.
- Insert sample records for at least 5 books and 3 borrowers.
- Retrieve book details borrowed by a specific borrower using nested queries.
- Count how many books each borrower has taken using aggregate functions.
- Create a view showing borrower name and books borrowed.

4. Perform the following operations:

- Create tables: Patient, Doctor, and Appointment.
- Insert sample data for at least 5 patients and 3 doctors.
- Retrieve patient names who have appointments with a specific doctor.
- Display total number of appointments per doctor.
- Create an index on Doctor_ID

5. Execute the following queries:

- Create tables: Customer, Product, and Orders.
- Insert sample data for 5 customers and 5 products.
- Display all products ordered by a specific customer using JOIN.
- Find customers who haven't placed any orders using NOT IN.
- Create a view that shows product name, quantity, and customer name.

6. Execute the following queries:

Given a single unnormalized `Sales` table (`book_id`, `book_title`, `author_name`, `customer_name`, `customer_address`, `purchase_date`):

- Convert it to 1NF.
- Further convert to 2NF.
- Normalize to 3NF and show the resulting tables.
- Explain primary and foreign keys in the resulting schema.

7. Perform the following operations:

- Create tables: `Teacher`, `Class`, `Subject`.
- Insert 5 records each.
- Display subject-wise list of teachers using `JOIN`.
- Count how many teachers are assigned per class.
- Create a view to show class name and total number of subjects taught.

8. Create a table `Account(Account_No, Customer_Name, Balance)` with appropriate data types and primary key constraint.

- Insert at least 3 sample records into the table.
- Write a PL/SQL block that:
 - Accepts an account number and withdrawal amount.
 - Checks if the account has sufficient balance.
 - If sufficient, deducts the amount and updates the balance.
 - If not, displays an appropriate message.
 - Includes exception handling for invalid account numbers and other errors.

PART B

9. Create a table `Bank_Account(Account_No, Customer_Name, Balance)` with suitable data types. Execute the following:

- Insert at least 3 records into the table.
- Write a PL/SQL block to simulate two transactions:
- Use `SAVEPOINT` to mark intermediate points.
- Use `ROLLBACK` to revert to a savepoint.
- Use `COMMIT` to finalize changes.
- Display the changes before and after rollback and commit to observe the effects.

10. Create a MongoDB collection named `Students` with fields such as `student_id`, `name`, `department`, and `marks`.

- Perform the following operations using appropriate MongoDB commands:
- Insert at least 5 student records.
- Update the marks of a student based on `student_id`.
- Delete a student record.
- Search for all students in a particular department.

11. Perform the following operations:

- Use the Students collection created earlier with fields including department and marks.
- Perform aggregation operations using MongoDB's aggregation pipeline:
- Count the number of students in each department.
- Calculate the average marks per department.

12. Perform the following operations:

- Create a sample .txt file containing plain English text.
- Upload the file to the Hadoop Distributed File System (HDFS).
- Run the default Hadoop **WordCount** program to process the uploaded file.
- Display and interpret the output that shows word frequencies.

13. Perform the following operations:

- Load a CSV file containing student information with fields such as student_id, name, GPA, and department using Spark SQL.
- Register the data as a temporary view or DataFrame.
- Write and execute queries to:
- Display all student records.
- Filter and show students with GPA > 8.
- Group students by department and display count.

14. Perform the following operations:

- Create two new users in the database with appropriate usernames and passwords.
- Grant SELECT and INSERT privileges on a table (e.g., Students) to one of the users.
- Log in as that user and perform operations to verify access.
- Revoke the privileges and attempt the same operations to observe the effect.
- Discuss the difference between **GRANT** and **REVOKE** operations.